



TASK 1

EMBEDDED SYSTEMS & IOT

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EMBEDDED SYSTEMS

An embedded system is a dedicated computer designed to perform one specific function within a larger machine or electrical system.

KEY CHARACTERISTICS

Single Task: It does only one specific job instead of running multiple applications.

Built-in Software: The program is loaded directly into the chip's permanent memory.

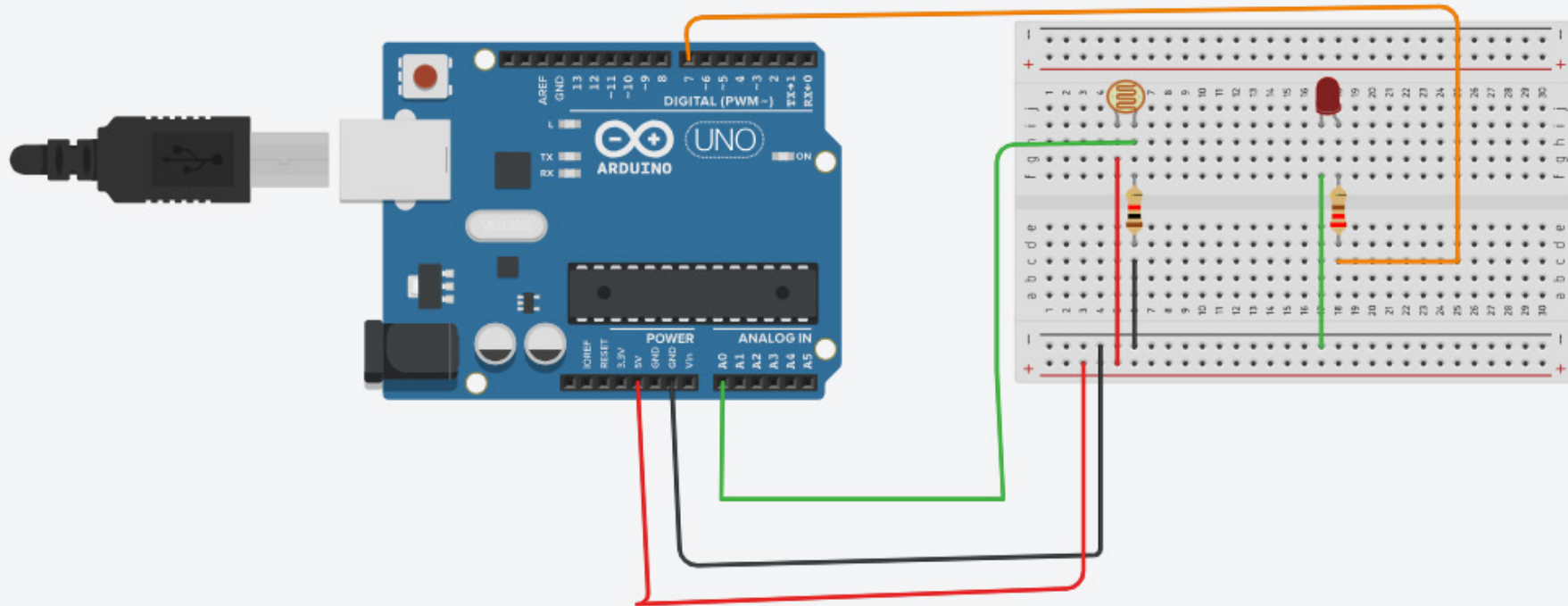
KEY CHARACTERISTICS

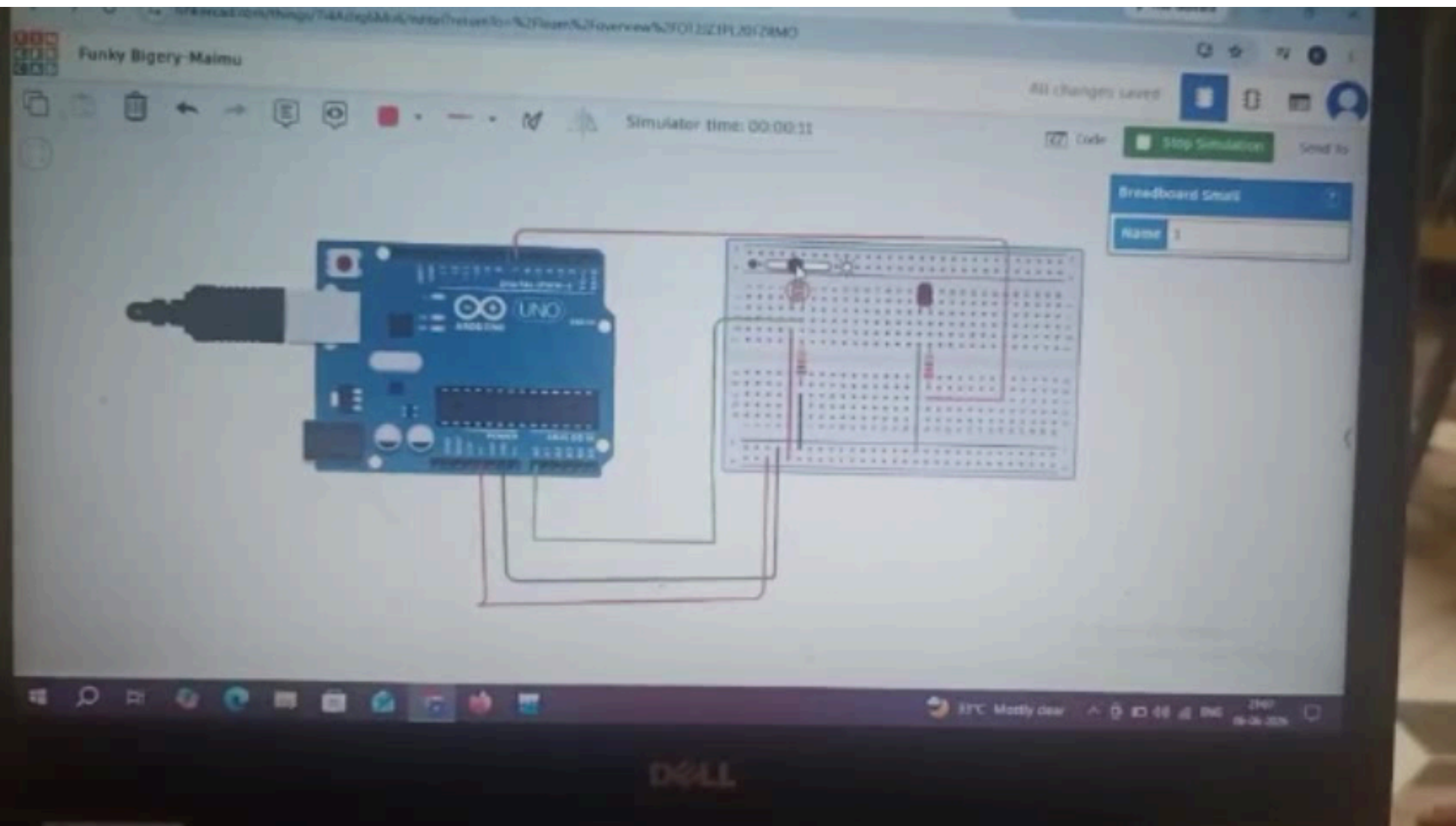
Real-time Response: It reacts to changes instantly with highly predictable timing.

Limited Resources: It operates with very small memory, low power, and tiny space.

Self-Running: It works automatically without human control and rarely crashes.

SMART LIGHTING SYSTEM





OUTPUT SIMULATION

```
const int ldrPin = A0; // LDR connected to analog pin A0  
const int ledPin = 7; // LED connected to digital pin 7
```

```
void setup() {  
  pinMode(ledPin, OUTPUT);  
  Serial.begin(9600); // For viewing sensor values in the monitor  
}
```

```
void loop() {  
  int sensorValue = analogRead(ldrPin); // Read the light value  
  Serial.println(sensorValue); // Print the value to Serial Monitor
```

```
  // Threshold is set to 500 (adjust this number depending on your lighting condition)  
  if (sensorValue < 500) {  
    digitalWrite(ledPin, HIGH); // Turn LED ON in the dark  
  } else {  
    digitalWrite(ledPin, LOW); // Turn LED OFF in the light  
  }
```

```
  delay(100); // Small delay to stabilize readings  
}
```


APPLICATIONS

Smart Homes

Street Lighting

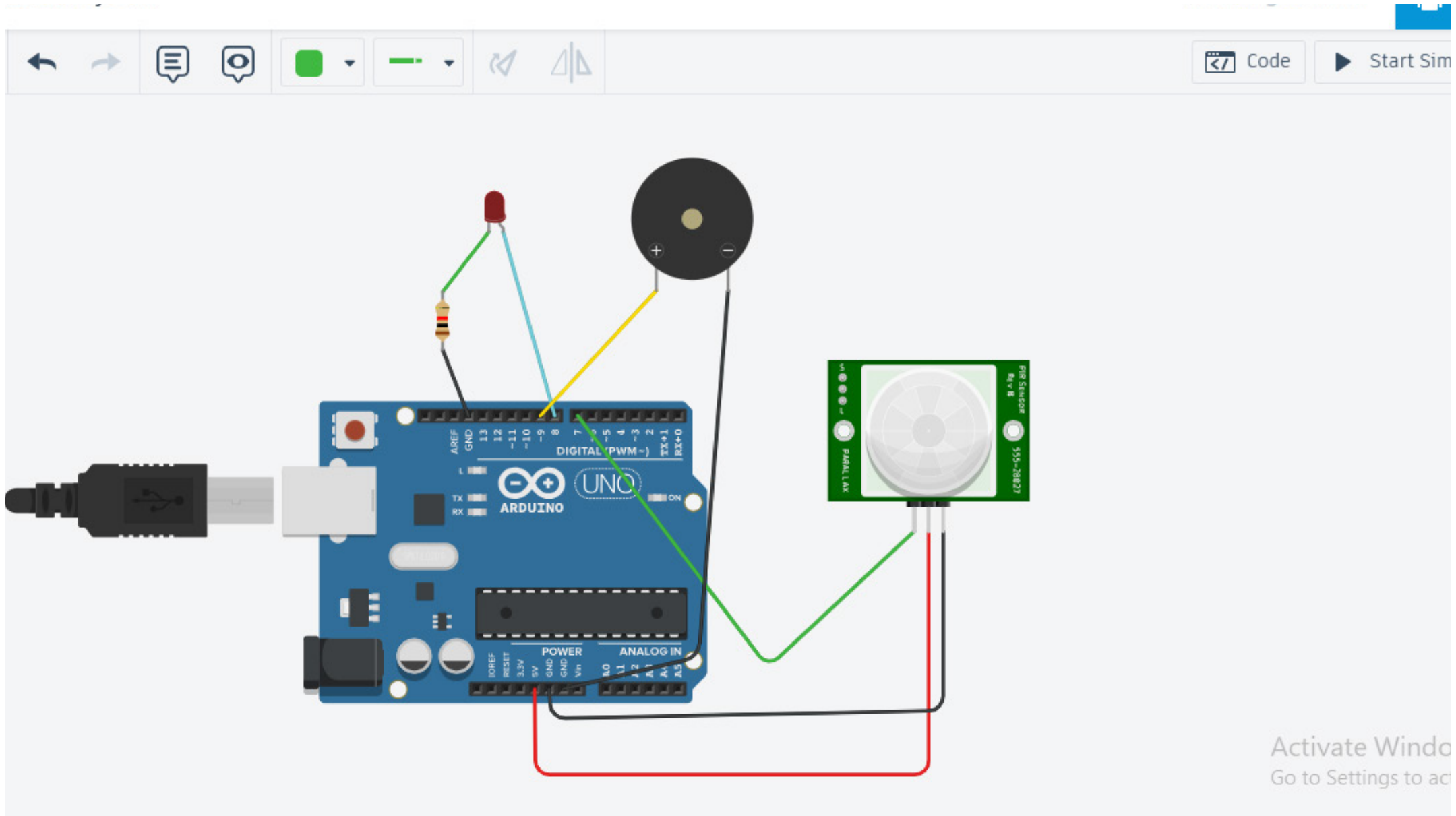
Office Buildings

Greenhouse Farming

Large Warehouses

Hospitals

MOTION DETECTION SYSTEM



Ask Gemini

arduino.com/thingy/TTGnG00zVS/edid/1e7um1o-NzFhmgp/6z7f6lydFW54t-ingenius-1e7m1o

Ask Google

Dazzling Robo

All changes saved

Simulator time: 00:00:07

Code

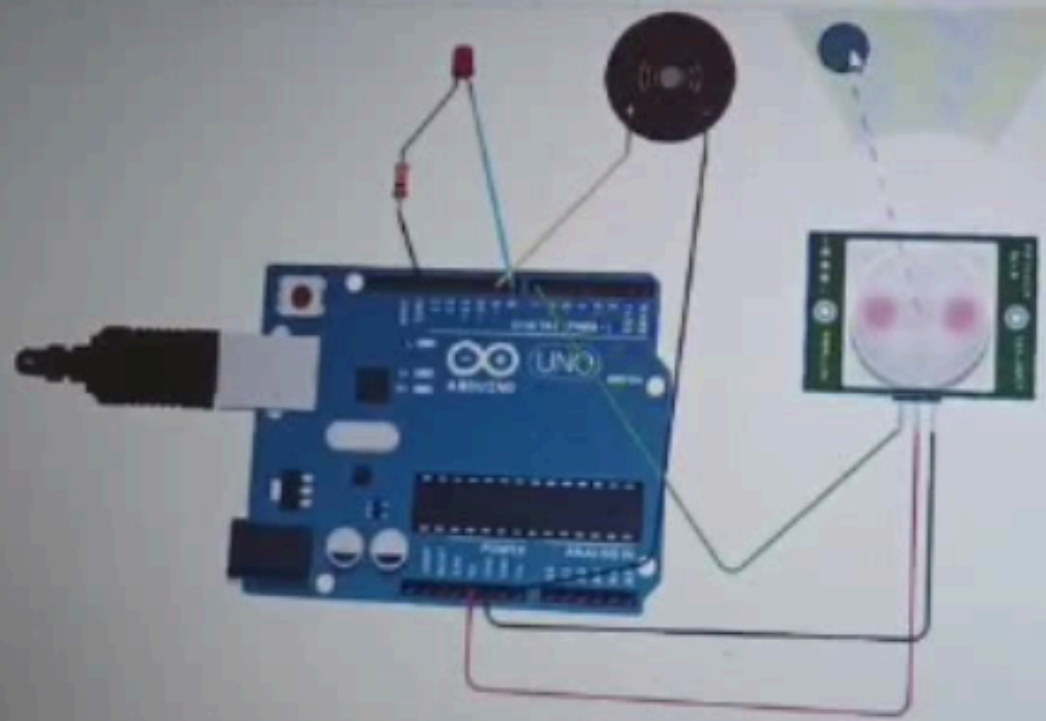
Stop Simulation

Send To

PIR Sensor

Name

1

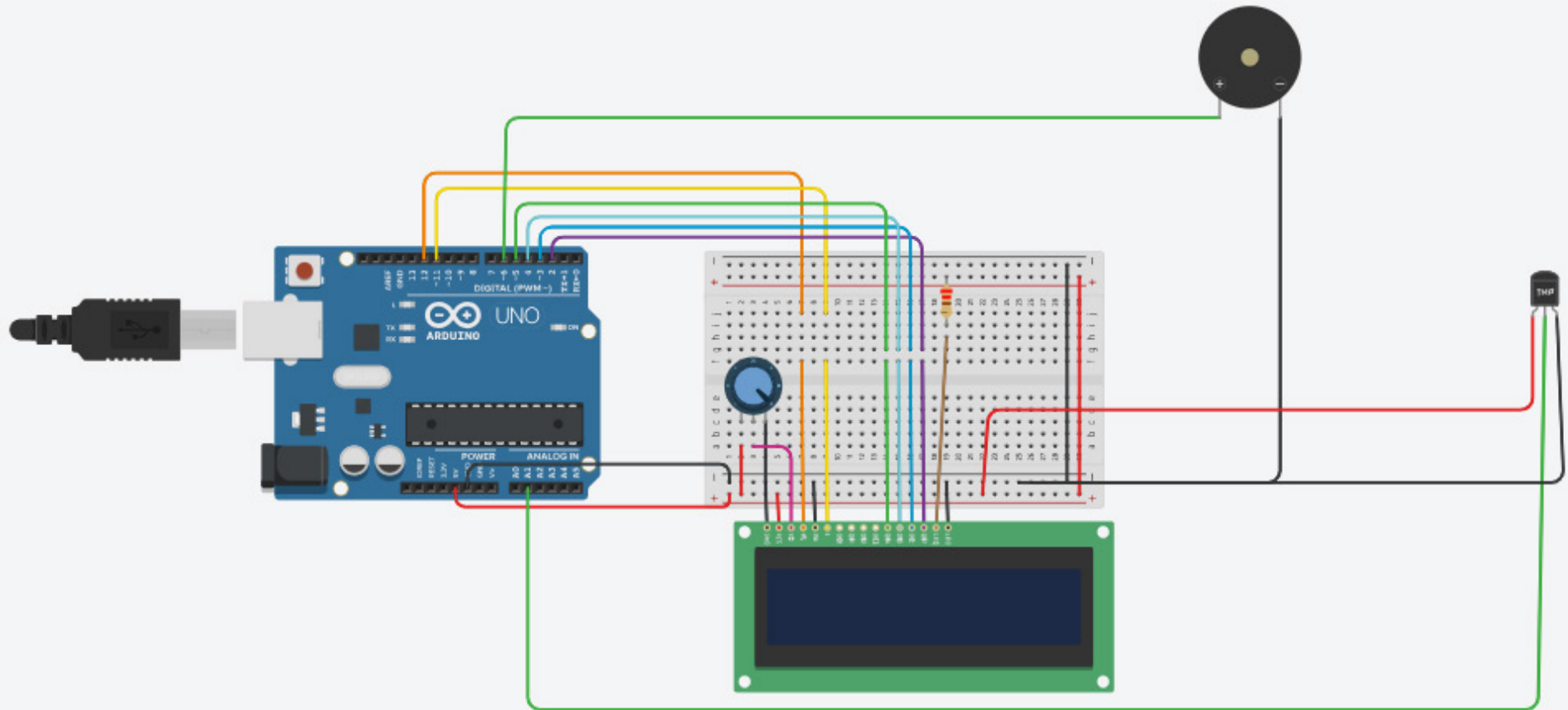


Activate Windows

Go to Settings to activate Windows.

```
void setup() {  
  pinMode(7, INPUT);  
  pinMode(8, OUTPUT);  
  pinMode(9, OUTPUT);  
}  
void loop()  
{  if (digitalRead(7) == HIGH)  
{  digitalWrite(8, HIGH);  
tone(9, 523, 1000); // play tone 60 (C5 = 523  
Hz)  }  
  
else  
  
{  digitalWrite(8, LOW);  
noTone(9);  }  
  
delay(1000); // Wait for 1000 millisecond(s)  
}
```

SMART TEMPERATURE MONITOR SYSTEM



```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
float a;
```

```
void setup() {
```

```
    lcd.begin(16, 2);
```

```
    pinMode(6,OUTPUT);
```

```
}
```

```
void loop() {
```

```
    a=analogRead(1);
```

```
    a=a*0.0048828125;
```

```
    a=(a-0.5)*100;
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print(a);
```

```
    lcd.print("C");
```

```
    if(a>50)
```

```
    {
```

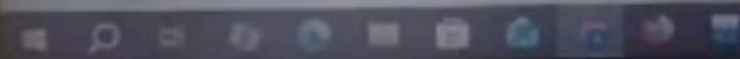
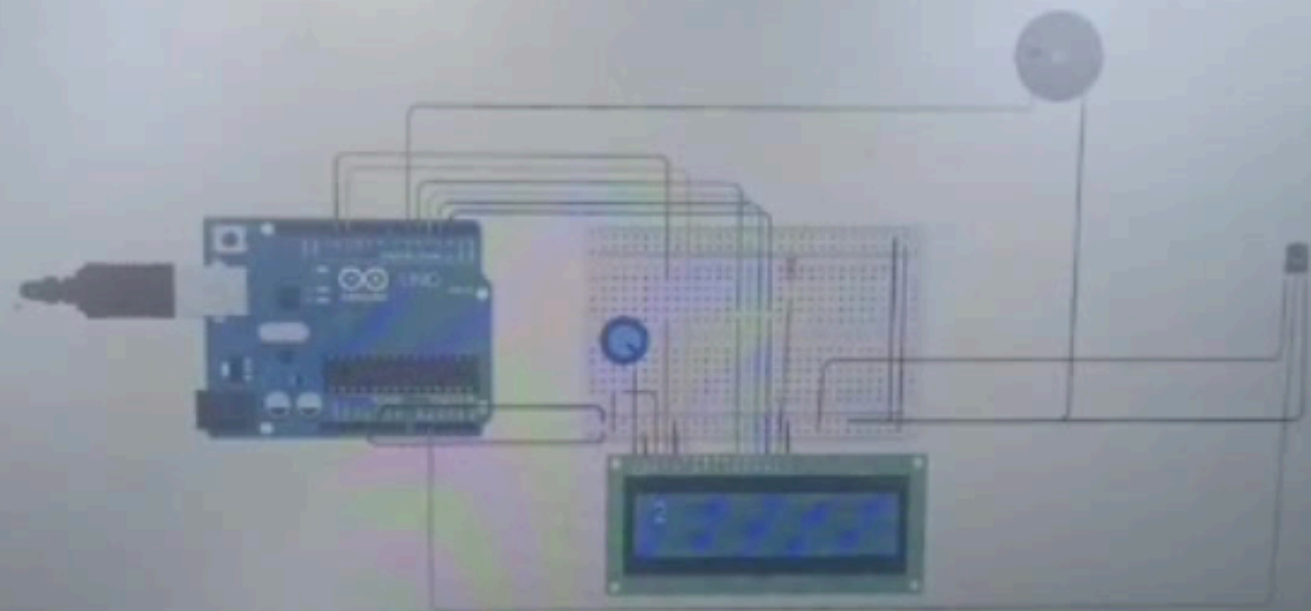
```
        digitalWrite(6,HIGH);
```

```
    }
```

```
    else
```

```
        digitalWrite(6,LOW);
```

```
}
```



DELL

CONCLUSION

EMBEDDED SYSTEM IS ALSO USED IN :

Smart Homes

Consumer Electronics

Automotive

Healthcare

Industrial Automation

Aerospace & Defense

Networking & Communication